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(54) **QUANTIFYING EFFECTS OF SEQUENCING
VARIANCE ON CLASSIFICATION**

(57) **ABSTRACT**

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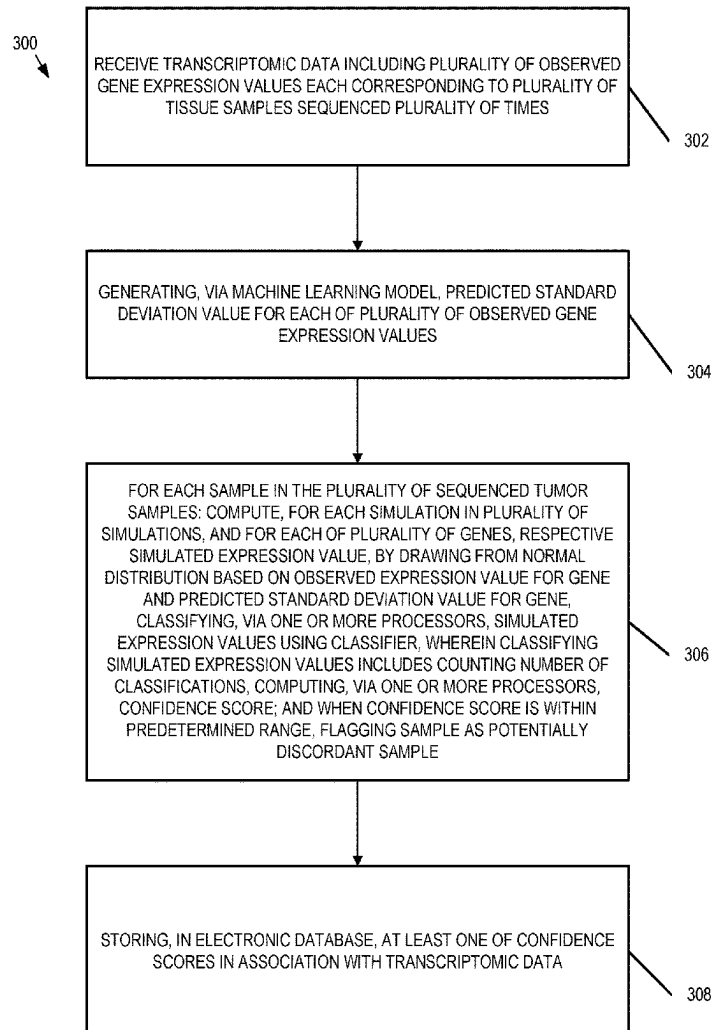
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A method includes receiving transcriptomic data; predicting standard deviation values for each gene expression values; for each sample: computing a simulated expression value, classifying the simulated expression values, computing confidence scores; and flagging the sample; and storing the confidence scores. A computing system includes a processor; and a memory having stored thereon instructions that when executed, cause the computing system to: receive transcriptomic data; predict standard deviation values for each gene expression values; for each sample: compute a simulated expression value, classify the simulated expression values, compute confidence scores; and flag the sample; and store the confidence scores. A computer-readable media includes non-transitory computer-readable instructions that, when executed, cause a computer to: receive transcriptomic data; predict standard deviation values for each gene expression values; for each sample: compute a simulated expression value, classify the simulated expression values, compute confidence scores; and flag the sample; and store the confidence scores.



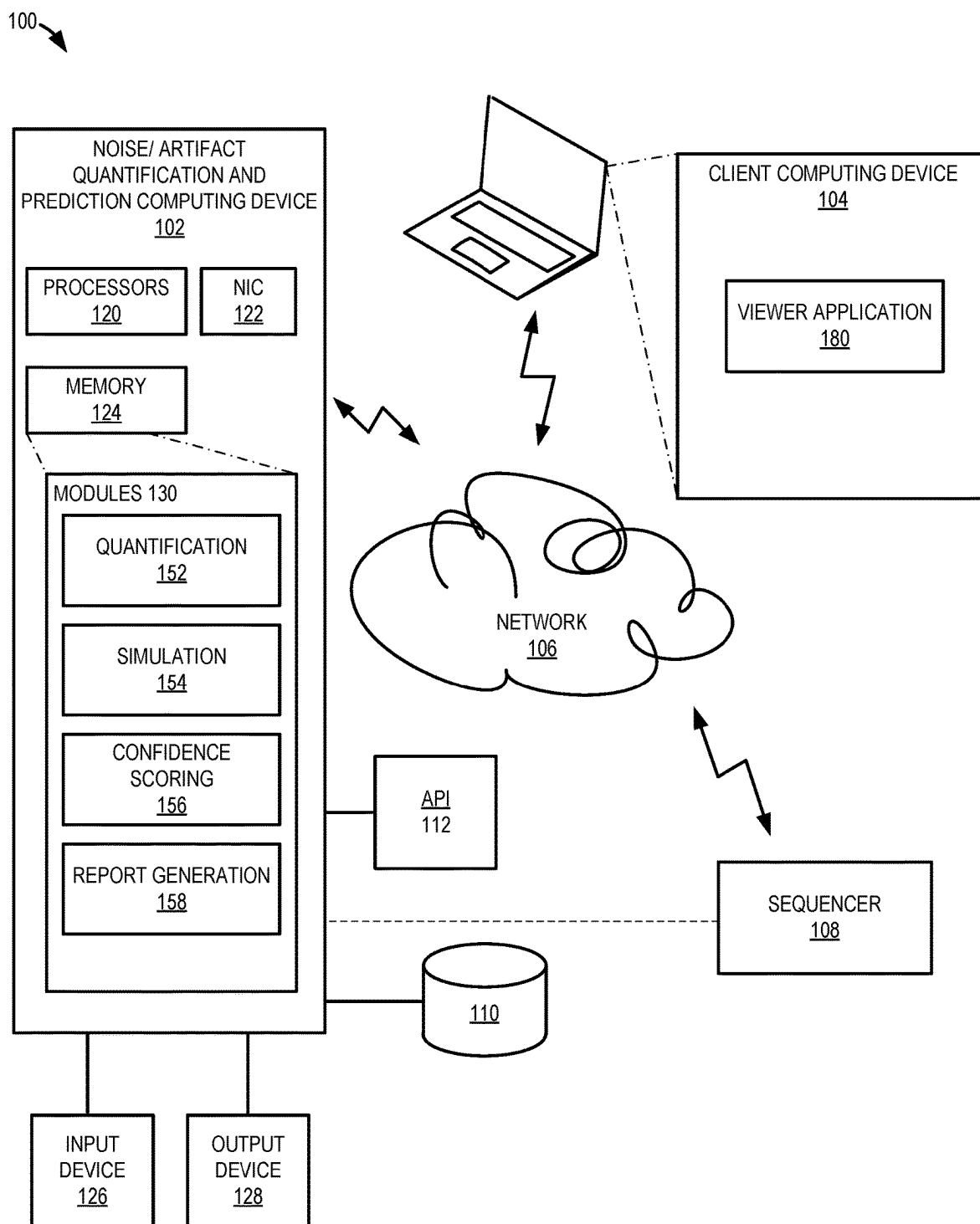


FIG. 1

200

“An indeterminate result occurs when the expression levels of gene pairs [or feature genes] used in the algorithm provide results that are within statistical margins of error between those defining the Group A and Group B subtypes; therefore the subtype classification cannot be reliably predicted. An indeterminate result is not due to poor sample quality or technical difficulties. Indeterminate results are expected to occur in about 7% of relevant patients.”

FIG. 2

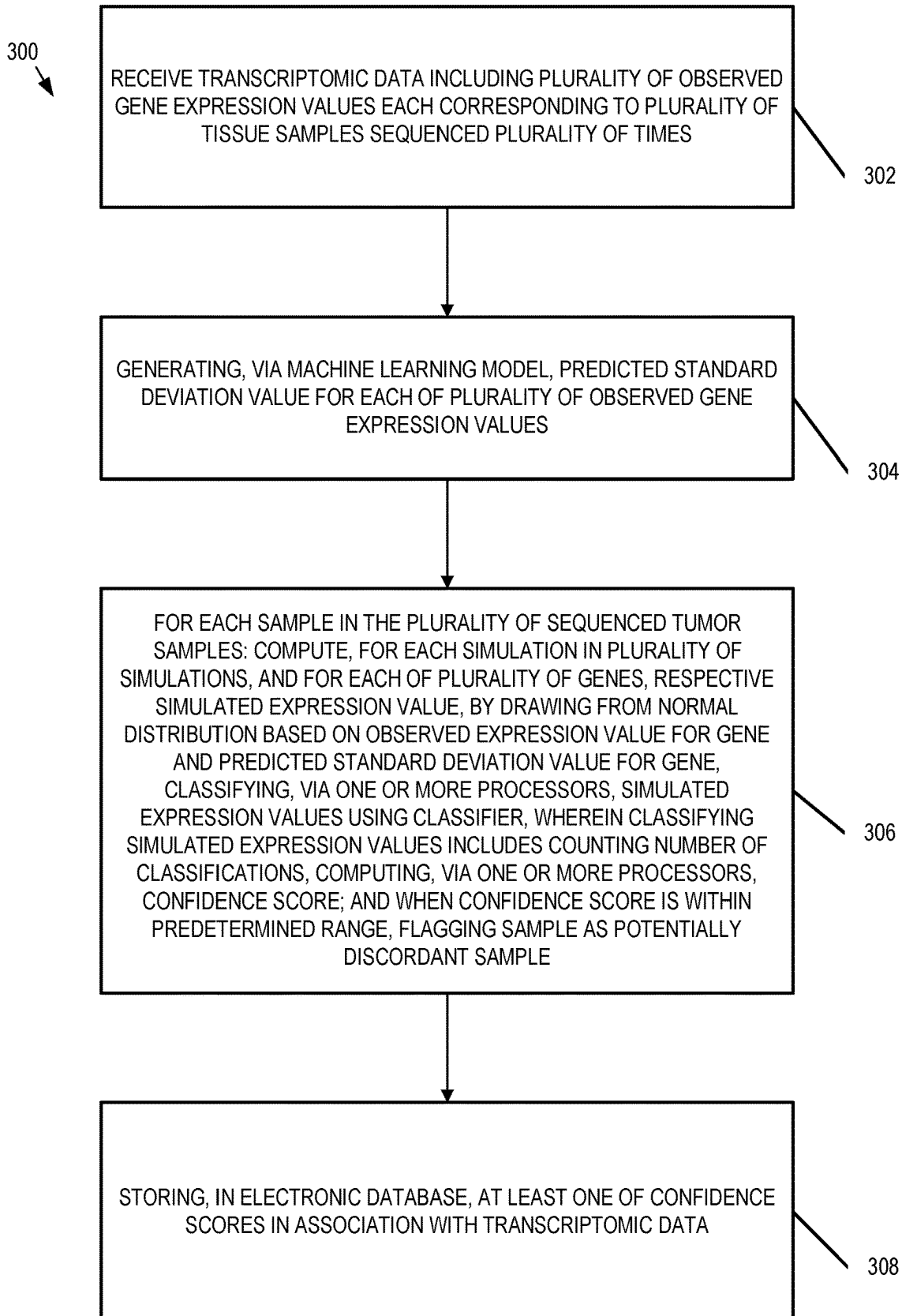


FIG. 3

QUANTIFYING EFFECTS OF SEQUENCING VARIANCE ON CLASSIFICATION

FIELD

[0001] The present disclosure is directed to methods and systems for quantifying effects of sequencing variance on classification, and more particularly, to techniques for performing role-based quality control checks on classification data.

BACKGROUND

[0002] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0003] Noise and artifacts are associated with gene expression levels due to the way gene expression levels are measured. For example, RNA expression quantification techniques (e.g., a sequencing machine, a microarray or another RNA expression quantification technique) introduce variance into the RNA-seq datasets that they generate. Specifically, sequencing a sample multiple times will produce multiple datasets, wherein a single gene common to the datasets can have different expression values in each respective dataset. For example, the multiple datasets may correspond to individual datasets generated by each run of the sample. Sometimes more than one sequencing run is required to obtain multiple datasets (one run can produce multiple datasets if two “copies”/aliquots/library preparations of the sample are run together on the sequencer, wherein each “copy” corresponds to a respective generated dataset).

[0004] This noise and artifact condition may lead to indeterminate results. For example, algorithms (e.g., a machine learning model, a classifier, etc.) that use continuous measures (e.g., gene expression values measured by next-generation sequencing RNA-seq) to perform a task (e.g., a classification-based patient binning/grouping task) may produce low-confidence or indeterminate results. In the classification example, expression levels of genes used in the algorithm may be within statistical margins of error of a number of classes (e.g., subtype A, subtype B, etc.). In such a case, a sample classified as subtype A may have RNA-seq values close to a threshold between two bins (classes), and if the sample were to be sequenced a second time, the variance introduced by the sequencer may cause the RNA-seq values to be different enough that the sample could be classified as subtype B.

[0005] This discordance and indeterminateness that is the result of sequencing noise and artifacts has been empirically observed, and is a problem in multiple contexts. First, indeterminate classification has been observed during validation of patient data that required multiple sequencing runs or reactions. This caused validation to fail, and required redesign of the testing. Second, indeterminate results have been observed in a clinical setting, preventing clinicians from selecting optimal therapy regimens. Communicating

discordant results to patients due to a lack of reproducible sequence-based testing has the potential to erode patient confidence.

[0006] Accordingly, there is an opportunity for improved platforms and technologies for quantifying sequence variance on classification, to quantify effects of noise and artifacts on sequencing variance.

SUMMARY

[0007] In an aspect, a computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample includes (i) receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) computing, via one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample; and (iv) storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0008] In another aspect, a computing system includes one or more processors; and one or more memories, having stored thereon instructions that when executed, cause the computing system to: (i) receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) compute, via the one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and (iv) store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0009] In yet another aspect, a computer-readable media includes a set of non-transitory computer-readable instructions that, when executed, cause a computer to: (i) receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generate, via a machine

learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) compute, via the one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and (iv) store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The figures described below depict various aspects of the system and methods disclosed herein. It should be understood that each figure depicts an example of aspects of the present systems and methods.

[0011] FIG. 1 depicts an exemplary computing system for quantifying sequencer variance, according to some aspects.

[0012] FIG. 2 depicts exemplary no-call language that may be included in a report comprising indeterminate results, according to some aspects.

[0013] FIG. 3 depicts an exemplary computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, according to some aspects.

DETAILED DESCRIPTION

Overview

[0014] The present techniques are directed to methods and systems for quantifying sampling noise during RNA expression quantification, and effects on models such as classifiers.

[0015] Empirically, “flips” between patient classes/groups/bins (for example, where more than one RNA-sequencing dataset is associated with the same sample, and the datasets are not all classified into the same group or bin) have been observed during algorithm validation. These discordant classifier results can be confusing for the clinician treating the patient, especially if each patient group is associated with a distinct therapy recommendation. In one example, the cause of these classification flips have been narrowed down to noise and artifacts in data generated during next-generation sequencing. Thus, the present techniques, which enable quantifying and predicting such flips or discordant results, provide a significant improvement to sequence data-based prediction techniques, by notifying researchers/clinicians of such potential indeterminate results, and by performing automatic mitigation.

[0016] In clinical cases (for example, where a sequenced sample is associated with a patient or organoid and the classifier result is reported to a clinician, especially to aid the clinician in selecting a therapy for a patient), it is useful to know when a sample might produce a discordant result on re-run (i.e., an indeterminate result), so that the clinician can receive accurate information to best select a therapy regimen for the patient, especially if subtype A is matched with

therapy A, and subtype B is matched with therapy B. It is useful to flag these indeterminate samples when reporting results to individual patients or clinicians to ensure a high degree of patient-level reproducibility. Of course, the subtypes A and B are mere examples (any number of classes may be chosen, as in a multi-stage model).

[0017] The present techniques may use proprietary estimates of confidence in gene expression values (e.g., for the probe enhanced whole-exome capture RNA-seq platform) to predict the likelihood of a sample producing discordant subtype results upon a re-run. Herein, samples with a high likelihood of showing discordant results upon multiple re-runs are considered low-confidence or indeterminate samples. In one example, samples that fell below an 85% confidence threshold may be classified as low-confidence and reported to patients (via their clinicians) as “no-calls”. In another example, those samples may be reported as indeterminate, equivocal, etc. Conversely, samples with confidence scores greater than or equal to 85% may be classified as high confidence and thus reportable to patients/clinicians. In other embodiments, the threshold for reporting may be 60%, 65%, 70%, 75%, 80%, 90%, 95%, etc.

[0018] The present techniques may include a simulation-based approach to calculate subtype confidence scores for each sample based on its proximity to the threshold between two subtypes. In some aspects, the present simulation-based approaches may resemble those described in “Real-world data validation of the PurlST pancreatic ductal adenocarcinoma gene expression classifier and its prognostic implications,” Wenric, et al., medRxiv 2023.02.23.23286356; doi: <https://doi.org/10.1101/2023.02.23.23286356>, hereby incorporated by reference in its entirety for all purposes.

[0019] Samples with confidence scores below a threshold may be flagged as “no-calls” and not assigned a subtype; further, in some aspects, these no-call samples may be censored from analytical and clinical validations of the algorithm and may not have an associated subtype reported to clinicians or patients.

[0020] In one aspect, the present techniques quantify confidence by using reference samples (e.g., universal human reference RNA (UHR)). The present techniques may compare noise from many (e.g., thousands or more) sequencing runs to parameterize noise specific to the sequencers used during those sequencing runs. For example, a standard deviation of UHR 20 k genes may be used to set the coefficients (this is sequencer dependent). This enables the present techniques to develop a dynamic confidence threshold for each gene. Examples of sequencers include Illumina HiSeq 4000 and Illumina NovaSeq 6000. Methods of next-generation sequencing for use in accordance with methods described herein are disclosed in Shendure 2008 Nat. Biotechnology 26:1135-1145 and Fullwood et al. 2009 Genome Res. 19:521-532, which are each hereby incorporated by reference in their respective entireties, for all purposes. Next generation sequencing methods well known in the art include synthesis technology (Illumina), pyrosequencing (454 Life Sciences), ion semiconductor technology (Ion Torrent sequencing), single-molecule real-time sequencing (PacBio Biosciences), sequencing by ligation (SOLID sequencing), nanopore sequencing (Oxford Nanopore Technologies), or paired-end sequencing. In some embodiments, massively parallel sequencing is performed using sequencing-by-synthesis with reversible dye terminators.

[0021] The present techniques are applicable to any RNA expression model that is used in a discretized fashion, and any sequencer that generates data (e.g., a set of quantified RNA expression values) that can be compared, enabling variance to be computed. Values whose variance may be quantified include gene expression levels, protein expression levels, enzyme activity, etc. However, the present techniques may be applicable to other continuous values.

[0022] In some aspects, the present techniques may be offered to another party, for example via a platform-as-a-service (PaaS) offering, software-as-a-service (SaaS), a white labeled service, etc. Such other parties may include pharmaceutical companies, biotechnical companies, contract research organizations, startups (e.g., personalized medicine firms), healthcare analytics companies, software companies, pharmacogenomics companies, healthcare consultancies, non-profit research organizations, etc.

Exemplary Computing Environments

[0023] FIG. 1 illustrates an exemplary computing environment 100 for performing the present techniques, according to some aspects. The environment 100 may include computing resources for quantifying noise and other artifacts, and for predicting whether the noise/artifacts will result in classification indeterminacy, and for further processing/computations based on such predictions.

[0024] The computing environment 100 may include a noise/artifact quantification and prediction computing device 102, a client computing device 104, an electronic network 106, a sequencer system 108 and an electronic database 110. The prediction computing device 102 may include an application programming interface 112 that enables programmatic access to the prediction computing device 102. The components of the computing environment 100 may be communicatively connected to one another via the electronic network 106, in some aspects. Each will now be described in greater detail.

[0025] The noise/artifact quantification and prediction computing device 102 may implement, inter alia, receipt/storage/preprocessing of quantified data (e.g., from the sequencer 108), simulation-based algorithms, processing of quantified data, predictive tasks and related operations. In some aspects, the noise/artifact quantification and prediction computing device 102 may be implemented as one or more computing devices (e.g., one or more servers, one or more laptops, one or more mobile computing devices, one or more tablets, one or more wearable devices, one or more cloud-computing virtual instances, etc.). The noise/artifact quantification and prediction computing device 102 may include one or more processors 120, one or more network interface controllers 122, one or more memories 124, an input device 126 and an output device 128.

[0026] In some aspects, the one or more processors 120 may include one or more central processing units, one or more graphics processing units, one or more field-programmable gate arrays, one or more application-specific integrated circuits, one or more tensor processing units, one or more digital signal processors, one or more neural processing units, one or more RISC-V processors, one or more coprocessors, one or more specialized processors/accelerators for artificial intelligence or machine learning-specific applications, one or more microcontrollers, etc.

[0027] The quantitative measures prediction computing device 102 may include one or more network interface

controllers 122, such as Ethernet network interface controllers, wireless network interface controllers, etc. The network interface controllers 122 may include advanced features, in some aspects, such as hardware acceleration, specialized networking protocols, etc.

[0028] The memories 124 of the quantitative measures prediction computing device 102 may include volatile and/or non-volatile storage media. For example, the memories 124 may include one or more random access memories, one or more read-only memories, one or more cache memories, one or more hard disk drives, one or more solid-state drives, one or more non-volatile memory express, one or more optical drives, one or more universal serial bus flash drives, one or more external hard drives, one or more network-attached storage devices, one or more cloud storage instances, one or more tape drives, etc.

[0029] The memories 124 may have stored thereon one or more modules 130, for example, as one or more sets of computer-executable instructions. In some aspects, the modules 130 may include additional storage, such as one or more operating systems (e.g., Microsoft Windows, GNU/Linux, Mac OSX, etc.). The operating systems may be configured to run the modules 130 during operation of the noise/artifact quantification and prediction computing device 102—for example, the modules 130 may include additional modules and/or services for receiving and processing quantitative data. The modules 130 may be implemented using any suitable computer programming language(s) (e.g., Python, JavaScript, C, C++, Rust, C#, Swift, Java, Go, LISP, Ruby, Fortran, etc.).

[0030] The modules 130 may include a quantification module 152, a simulation module 154, a confidence scoring module 156 and a report generation module 158. In some aspects, more or fewer modules 130 may be included. The modules 130 may be configured to communicate with one another (e.g., via inter-process communication, via a bus, message queue, sockets, etc.).

[0031] The quantification module 152 may collect data comprising continuous data values during an experimentation process, such as during the execution of one or more sequencers operated by the sequencer system 108. The quantification module 152 may include sets of computer-executable instructions for collecting and storing the continuous data values, for example, as a data stream. The quantification module 152 may store the collected continuous data values in the electronic database 110. The quantification module 152 may include instructions for pre-processing the continuous data values prior to storing it, and for associating the continuous data values with other data or metadata (e.g., sample run date/time, batch number, sample identifier, etc.).

Exemplary Computer-Implemented Confidence Score Generation

[0032] The simulation module 154 may include computer-executable instructions for processing the collected continuous data values to generate confidence scores. For example, the simulation module 154 may include one or more algorithms for generating a predicted standard deviations for observed values. For example, the simulation module 154 may perform linear model computations.

[0033] In the case of the continuous data values corresponding to expression values for one or more features (e.g.,

one or more genes), the simulation module **154** may generate a predicted standard deviation by computing a linear model defined as

$$\text{Predicted_standard_deviation} \sim a * \sqrt{\log_{10}(\text{observed_expression})} - b,$$

where constants *a* and *b* are determined by fitting the model to a dataset comprising a number (e.g., 1000) of replicate runs of a single sample (e.g., a universal human reference sample) on an RNA-Seq pipeline. For example, for one particular sequencing machine model, the constants *a* and *b* may be -0.08344563 and 0.2120142 , respectively. The simulation module **154** may compute the standard deviation of each observed expression value for each algorithm feature (e.g., gene) using the linear model. The linear model may predict standard deviation based on mean expression alone and may be agnostic to gene identity.

[0034] In some aspects, the one or more features may be those of PurIST, a k-top scoring pairs (k-TSP) gene signature model that assesses rank-based expression levels of 16 genes sorted into 8 gene pairs from a single sample to estimate the likelihood of that sample being basal. PurIST is described in “Purity Independent Subtyping of Tumors (PurIST), A Clinically Robust, Single-Sample Classifier for Tumor Subtyping in Pancreatic Cancer”; Clinical Cancer Research: An Official Journal of the American Association for Cancer; Research 26 (1): 82-92. Rashid et al. 2020, hereby incorporated by reference in its entirety, for all purposes. It should be appreciated that other subtyping algorithms (e.g., k-TSP based algorithms or other algorithms, single-gene models, and subtyping models including those derived from hierarchical clustering, consensus clustering, NMF, k-means, kNN and any other subtyping model making use of gene expression data to classify and label samples) may benefit from the present techniques, in addition to, or alternatively from PurIST. Generally, any model that uses any number of continuous values subject to measurement error (e.g., gene expression values) to determine discrete subtypes may be aided by the present techniques.

[0035] Other classifiers that may be used include those discussed in U.S. Pat. No. 11,527,323 entitled “Systems and methods for multi-label cancer classification,” filed on May 12, 2020 and hereby incorporated by reference herein in its entirety; and those discussed in U.S. Patent Application Publication No. 20220154284, entitled “Determination of cytotoxic gene signature and associated systems and methods for response prediction and treatment,” filed on Nov. 19, 2021, and incorporated hereby in reference in its entirety. In some aspects, the classifiers discussed in U.S. patent application Ser. No. 17/738,935, entitled “Techniques for resolving discordant her2 status in pathology images for diagnosing breast cancer” and filed on May 6, 2022, hereby incorporated by reference in its entirety, may be used. The present techniques may be extended to any subtyping algorithm, and PurIST is merely one example. For example, in some aspects, the present techniques may be used to quantify variance in gene co-expression analysis, differential expression analysis, gene network inference techniques, etc.

[0036] Next, the simulation module **154** may simulate a number of classification values, by drawing a number (e.g., 16) of gene log-transformed expression values from a number (e.g., 16) of normal distributions defined using the

observed log-transformed expression values as their means along with their respective predicted standard deviations. The simulation module **154** may then apply the classifier to the simulated classifications to determine a simulated predicted classification. This process may be performed numerous times, for example, from simulations 1 . . . N, a new set of simulated expression values may be selected from the normal distribution.

[0037] The simulation module **154** may store the simulated expression values and the simulated predicted classification in the electronic database **110**, in some aspects. The simulation module **154** may also transmit the simulated expression values and the simulated predicted classification to another element of the environment **100**, or beyond, in some aspects. Next, the simulation module **154** may repeat the simulation procedure a number (e.g., 10,000 or more) of times, and the frequencies of predicted subtypes may be counted.

[0038] The confidence scoring module **156** may include computer-executable instructions for computing one or more confidence scores. For example, the module **156** may compute one or more confidence scores by determining a ratio of each given classification to a number of total simulations. For example, if the simulation is performed by the simulation module **154** for a given sample is 10,000 times, and 9000 of the classifications classify the sample in a bin X, then the score would be 0.90 for bin X. As noted, the classifier may be a binary classifier (in which the majority of classifications may constitute the score) or a multi classifier, in which a sample is classified into two or more bins.

[0039] Generally, for a binary classifier embodiment, a score of 0.50 is a very low confidence score, being equivalent to a coin flip, while a 1.0 score is a very high confidence score, with zero discordance expected. The confidence scoring module **156** may store the generated confidence score(s) for a given sample in the electronic database **110**, in some aspects. The confidence scoring module **156** may provide the one or more confidence scores to one or more other elements of the environment **100**, such as the report generation module **158**.

Exemplary Computer-Implemented Rules Engine

[0040] The present techniques may establish a confidence score cutoff, wherein any confidence score less than that score is considered a “no call,” meaning that the confidence level is below a predetermined threshold (e.g., 0.85). In some aspects, a no call confidence score may be used by the report generation module **158** to censor or abort communication to patients or others about corresponding classifications. The report generation module **158** may also allow “no call” classifications to be included in reports, but only with disclaimer language, “low confidence” flags, or other visual/graphical indicia, to advise the reader/viewer accordingly as to the potentially discordant status of the classification.

[0041] In some aspects, the report generation module **158** may generate static reports that include predictions regarding the confidence of classifications of patient data, as shown in FIG. 2. For example, these static reports may take the form of text documents, digital presentations, JSON documents, PDF documents, word processing documents, etc.

[0042] In some aspects, the report generation module **158** may include a rules engine for programmatically generating filtered sets of classification results based on confidence scores. For example, the electronic database **110** may

include a vector or table of information corresponding to a sequencing run of a patient's tissue. In that case, the rows of the table may each include a respective gene name (e.g., as column 1) and an associated respective expression level value (e.g., column 2). The report generation module 158 may process that vector of information and perform its classification prediction routines as discussed above, storing the results in the electronic database 110.

[0043] Later, the prediction computing device 102 may receive, e.g., via the viewer application 180, an electronic request (e.g., an HTTP POST or GET request) for classification results. The 102 may inspect the request to determine the identity of the requestor. In some aspects, the prediction computing device 102 may include a role module (not depicted) that accepts the identity of the requestor (e.g., as a session identifier, a login identifier, an authentication credential, etc.) and determines, based on that identity identifier, a role of the user (e.g., patient, clinician, research, etc.).

[0044] The rules engine of the report generation module 158 may include one or more sets of computer-executable instructions specifying which predictions the role is entitled to. The report generation module 158 may generate a query specific to the role, and issue that query against the database 110, to retrieve only those results that the user is entitled to, based on the user's role. The report generation module 158 may then generate an electronic response (e.g., an HTTP response) to the electronic request, and transmit the filtered results to the user. In some aspects, the report generation module 158 may include "no call" classification results in this response, but only after additional rules in the rules engine annotate those results as "no call" results using a flag or other indicia, as discussed above. In some aspects, a user (e.g., a clinician) may set a preference via the viewer application 180 that affects the way that "no call" results are identified in data sets.

[0045] The present reporting techniques represent an advantageous improvement over conventional classification confidence techniques, especially because the environment 100 can be plugged into a larger computational pipeline, to provide a completely automated solution for annotation and sharing of clinical results with no-call annotations or filtering. For example, instead of the user-initiated request/response interaction discussed above, in some aspects, the application programming interface 112 may receive authenticated requests from one or more scheduled computer programs (e.g., a script that runs periodically on a remote computing device). Such script programs may be configured to have a suitable role, so that the information they receive in responses from the application programming interface 112 is appropriately filtered/annotated.

[0046] The report generation module 158 may include computer-executable instructions for generating machine-readable results via the application programming interface 112. For example, the application programming interface 112 may generate a structured output (e.g., results encoded in JavaScript Object Notation, XML, etc.), so that a calling client of the application programming interface 112 can easily extract data from the results, including data corresponding to the confidence score of the data.

[0047] The client computing device 104 may be accessed by a user to view results generated by the prediction computing device 102. For example, a user may access a mobile device, laptop device, thin client, etc. embodied as the client

computing device 104 to view simulation and confidence scoring results, and/or reports, with respect to a sample whose continuous values were processed by the prediction computing device 102.

[0048] Information from the prediction computing device 102 may be transmitted via the network 106 (e.g., for display via the viewer application 180).

[0049] The electronic network 106 may communicatively couple the elements of the environment 100. The network 106 may include public network(s) such as the Internet, a private network such as a research institution or corporation private network, and/or any combination thereof. The network 106 may include a local area network (LAN), a wide area network (WAN), a cellular network, a satellite network, and/or other network infrastructure, whether wireless or wired.

[0050] In some aspects, the network 106 may be communicatively coupled to and/or part of a cloud-based platform (e.g., a cloud computing infrastructure). The network 106 may utilize communications protocols, including packet-based and/or datagram-based protocols such as Internet protocol, transmission control protocol, user datagram protocol, and/or other types of protocols. The network 106 may include one or more devices that facilitate network communications and/or form a hardware basis for the networks, such as one or more switches, one or more routers, one or more gateways, one or more access points (such as a wireless access point), one or more firewalls, one or more base stations, one or more repeaters, one or more backbone devices, etc.

[0051] The sequencer system 108 may include a next generation sequencer, such as a RNA sequencing whole exome capture transcriptome assay. As discussed in U.S. Pat. No. 11,043,283, entitled "Systems and methods for automating RNA expression calls in a cancer prediction pipeline," filed Dec. 4, 2020 and in U.S. Pat. No. 11,367,508, entitled "Systems and methods for detecting cellular pathway dysregulation in cancer," filed Aug. 14, 2020—both of which are hereby incorporated by reference in their respective entireties, for all purposes—in various embodiments, each transcriptome value set may be generated by processing a patient or tumor organoid sample through RNA whole exome next generation sequencing (NGS) to generate RNA sequencing data, and the RNA sequencing data may be processed by a bioinformatics pipeline to generate a RNA-seq expression profile for each sample. The patient sample may be a tissue sample or blood sample containing cancer cells. In more detail, RNA may be isolated from blood samples or tissue sections using commercially available reagents, for example, proteinase K, TURBO DNase-I, and/or RNA clean XP beads. The isolated RNA may be subjected to a quality control protocol to determine the concentration and/or quantity of the RNA molecules, including the use of a fluorescent dye and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. cDNA libraries may be prepared from the isolated RNA, purified, and selected for cDNA molecule size selection using commercially available reagents, for example Roche KAPA Hyper Beads. In another example, a New England Biolabs (NEB) kit may be used. cDNA library preparation may include the ligation of adapters onto the cDNA molecules. For example, UDI adapters, including Roche SeqCap dual end adapters, or UMI adapters (for example, full length or stubby Y adapters) may be ligated to

the cDNA molecules. The sequence of nucleotides in the adapters may be specific to a sample in order to distinguish between sequencing data obtained for different samples. In this example, adapters are nucleic acid molecules that may serve as barcodes to identify cDNA molecules according to the sample from which they were derived and/or to facilitate the next generation sequencing reaction and/or the downstream bioinformatics processing. cDNA libraries may be amplified and purified using reagents, for example, Axygen MAG PCR clean up beads. Then the concentration and/or quantity of the cDNA molecules may be quantified using a fluorescent dye and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. cDNA libraries may be pooled and treated with reagents to reduce off-target capture, for example Human COT-1 and/or IDT xGen Universal Blockers, before being dried in a vacufuge. Pools may then be resuspended in a hybridization mix, for example, IDT xGen Lockdown, and probes may be added to each pool, for example, IDT xGen Exome Research Panel v1.0 probes, IDT xGen Exome Research Panel v2.0 probes, other IDT probe panels, Roche probe panels, or other probes. Pools may be incubated in an incubator, PCR machine, water bath, or other temperature modulating device to allow probes to hybridize. Pools may then be processed with Streptavidin-coated beads, or another means for capturing hybridized cDNA-probe molecules, especially cDNA molecules representing exons of the human genome. In some embodiments, polyA capture may be used. Pools may be amplified and purified once more using commercially available reagents, for example, the KAPA HiFi Library Amplification kit and Axygen MAG PCR clean up beads, respectively. The cDNA library may be analyzed to determine the concentration or quantity of cDNA molecules, for example by using a fluorescent dye (for example, PicoGreen pool quantification) and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. The cDNA library may also be analyzed to determine the fragment size of cDNA molecules, which may be done through gel electrophoresis techniques and may include the use of a device such as a LabChip GX Touch. Pools may be cluster amplified using a kit (for example, Illumina Paired-end Cluster Kits with PhiX-spike in). In one example, the cDNA library preparation and/or whole exome capture steps may be performed with an automated system, using a liquid handling robot (for example, a SciClone NGSx). The amplification may be performed on a device, for example, an Illumina C-Bot2, and the resulting flow cell containing amplified target-captured cDNA libraries may be sequenced on a next generation sequencer, for example, an Illumina HiSeq 4000 or an Illumina NovaSeq 6000 to a unique on-target depth selected by the user, for example, 300x, 400x, 500x, 10,000x, etc. The next generation sequencer may generate a FASTQ file for each patient sample. Each FASTQ file contains reads that may be paired-end or single reads, and may be short-reads or long-reads, where each read shows one detected sequence of nucleotides in an mRNA molecule that was isolated from the patient sample, inferred by using the sequencer to detect the sequence of nucleotides contained in a cDNA molecule generated from the isolated mRNA molecules during library preparation. Each read in the FASTQ file is also associated with a quality rating. The quality rating may reflect the likelihood that an error occurred during the sequencing procedure that affected the associated read. The adapters may facilitate the binding

of the cDNA molecules to anchor oligonucleotide molecules on the sequencer flow cell and may serve as a seed for the sequencing process by providing a starting point for the sequencing reaction. If two or more patient samples are processed simultaneously on the same sequencer flow cell, reads from multiple patient samples may be contained in the same FASTQ file initially and then divided into a separate FASTQ file for each patient. A difference in the sequence of the adapters used for each patient sample could serve the purpose of a barcode to facilitate associating each read with the correct patient sample and placing it in the correct FASTQ file. Each FASTQ file may be processed by a bioinformatics pipeline. In various embodiments, the bioinformatics pipeline may filter FASTQ data. Filtering FASTQ data may include correcting sequencer errors and removing (trimming) low quality sequences or bases, adapter sequences, contaminations, chimeric reads, overrepresented sequences, biases caused by library preparation, amplification, or capture, and other errors. Entire reads, individual nucleotides, or multiple nucleotides that are likely to have errors may be discarded based on the quality rating associated with the read in the FASTQ file, the known error rate of the sequencer, and/or a comparison between each nucleotide in the read and one or more nucleotides in other reads that has been aligned to the same location in the reference genome. Filtering may be done in part or in its entirety by various software tools. FASTQ files may be analyzed for rapid assessment of quality control and reads, for example, by a sequencing data QC software such as AfterQC, Kraken, RNA-SeQC, FastQC, (see Illumina, BaseSpace Labs or [https://world wide web address illumina.com/products/by-type/informatics-products/basespace-sequence-hub/apps/fastqc.html](https://worldwide.web.address.illumina.com/products/by-type/informatics-products/basespace-sequence-hub/apps/fastqc.html)), or another similar software program. For paired-end reads, reads may be merged. For each FASTQ file, each read in the file may be aligned to the location in the reference genome having a sequence that best matches the sequence of nucleotides in the read. There are many software programs designed to align reads, for example, Bowtie, Burrows Wheeler Aligner (BWA), programs that use a Smith-Waterman algorithm, etc. Alignment may be directed using a reference genome (for example, GRCh38, hg38, GRCh37, other reference genomes developed by the Genome Reference Consortium, etc.) by comparing the nucleotide sequences in each read with portions of the nucleotide sequence in the reference genome to determine the portion of the reference genome sequence that is most likely to correspond to the sequence in the read. The alignment may take RNA splice sites into account. The alignment may generate a SAM file, which stores the locations of the start and end of each read in the reference genome and the coverage (number of reads) for each nucleotide in the reference genome. The SAM files may be converted to BAM files, BAM files may be sorted, and duplicate reads may be marked for deletion. In one example, kallisto software may be used for alignment and RNA read quantification (see Nicolas L Bray, Harold Pimentel, Pall Melsted and Lior Pachter, Near-optimal probabilistic RNA-seq quantification, *Nature Biotechnology* 34, 525-527 (2016), doi: 10.1038/nbt.3519). In an alternative embodiment, RNA read quantification may be conducted using another software, for example, Sailfish or Salmon (see Rob Patro, Stephen M. Mount, and Carl Kingsford (2014) Sailfish enables alignment-free isoform quantification from RNA-seq reads using lightweight algorithms. *Nature Bio-*

technology (doi: 10.1038/nbt.2862) or Patro, R., Duggal, G., Love, M. I., Irizarry, R. A., & Kingsford, C. (2017). Salmon provides fast and bias-aware quantification of transcript expression. *Nature Methods*, incorporated by reference herein in its entirety for all purposes). These RNA-seq quantification methods may not require alignment. There are many software packages that may be used for normalization, quantitative analysis, and differential expression analysis of RNA-seq data. For each gene, the raw RNA read count for a given gene may be calculated. The raw read counts may be saved in a tabular file for each sample, where columns represent genes and each entry represents the raw RNA read count for that gene. In one example, kallisto alignment software calculates raw RNA read counts as a sum of the probability, for each read that the read aligns to the gene. Raw counts are therefore not integers in this example. Raw RNA read counts may then be normalized to correct for GC content and gene length, for example, using full quantile normalization and adjusted for sequencing depth, for example, using the size factor method. In one example, RNA read count normalization is conducted according to the methods disclosed in U.S. patent application Ser. No. 16/581,706 or PCT19/52801, titled *Methods of Normalizing and Correcting RNA Expression Data* and filed Sep. 24, 2019, incorporated by reference herein in their respective entireties. The rationale for normalization is the number of copies of each cDNA molecule in the sequencer may not reflect the distribution of mRNA molecules in the patient sample. For example, during library preparation, amplification, and capture steps, certain portions of mRNA molecules may be over or under-represented due to artifacts that arise during various aspects of priming of reverse transcription caused by random hexamers, amplification (PCR enrichment), rRNA depletion, and probe binding and errors produced during sequencing that may be due to the GC content, read length, gene length, and other characteristics of sequences in each nucleic acid molecule. Each raw RNA read count for each gene may be adjusted to eliminate or reduce over- or under-representation caused by any biases or artifacts of NGS sequencing protocols. Normalized RNA read counts may be saved in a tabular file for each sample, where columns represent genes and each entry represents the normalized RNA read count for that gene. The electronic database **110** may include one or more suitable electronic databases for storing and retrieving data, such as relational database (e.g., MySQL databases, Oracle databases, Microsoft SQL Server databases, PostgreSQL databases, etc.). The electronic database **110** may be a NoSQL database, such as a key-value store, a graph database, a document store, etc. The electronic database **110** may be an object-oriented database, a hierarchical database, a spatial database, a time-series database, an in-memory database, etc. In some aspects, some or all of the electronic database **110** may be distributed.

[0052] In operation, one or more sequencer runs may be performed using the sequencer **108**, either by the company operating/controlling the environment **100** or by another party. The results of the sequencer are received by the prediction computing device **102**, which performs quantification **152** of the results, and optionally, storage of the quantified results. Next, the simulation module **154** processes the results, to generate confidence scores for the results. The confidence scores may be stored in association with the stored quantified results in the database **110**, for

example. Next the prediction computing device **102** receives a request for data. The prediction computing device **102** provides the data, after filtering and/or annotating the data by applying rules to parameters included in the request for data. Overall, the process results in a quality control check that prevents clinicians from misinterpreting low confidence data, and prevents patients from losing confidence in the sequencing process due to the potential for passing along indeterminate results-which is advantageously largely eliminated using the present techniques, because the quality control check is performed before data is ever provided to patients.

Exemplary Computer-Implemented Report Generation

[0053] FIG. 2 depicts exemplary no-call language **200** that may be generated by the rules engine of the report generation module **158**, in some aspects. For example, the no-call language of FIG. 2 may be the language generated when the rules engine determines, via the application programming interface **112** that the requestor is a patient computing device, or that the user making the request is logged in and has a role of patient.

[0054] The various embodiments described above can be combined to provide further embodiments. All U.S. patents, U.S. patent application publications, U.S. patent application, foreign patents, foreign patent application and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified if necessary to employ concepts of the various patents, applications, and publications to provide yet further embodiments.

[0055] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Exemplary Computer-Implemented Method Aspects

[0056] FIG. 3 depicts an exemplary computer-implemented quality control method **300** for predicting whether a continuous value classifier will generate discordant classifications of a sample, according to some aspects. The method may be performed by one or more of the components of environment **100** of FIG. 1, in some aspects.

[0057] The method **300** may include receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times (block **302**). The samples may have originated from various sources, including but not limited to, tissue samples, commercially available reference RNA, manufactured RNA molecules, etc. The method **300** may include generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values (block **304**). For example, the machine learning model may be a Bayesian model, a supervised machine learning model or an unsupervised machine learning model.

[0058] The method 300 may include for each sample in the plurality of samples (e.g., sequenced tumor samples): (i) computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (ii) classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (iii) computing, via one or more processors, a confidence score; and (iv) when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample (block 306).

[0059] The method 300 may include storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data (block 308).

[0060] In some aspects, the method 300 may include generating, via one or more processors, a static digital report including at least one classification result. In some aspects, generating the static digital report including the at least the one classification result may include censoring a no-call classification result.

[0061] In some aspects, the method 300 may include receiving, via an application programming interface, a request for sequence data (e.g., RNA sequence data); processing, via one or more processors, the request using a rules engine to determine an identity of the sender of the request; determining, via one or more processors, a role of the identity of the sender of the request; configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and/or transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request. In some aspects, generating the modified sequence data by processing the sequence data using the configured rules engine may generate censored results when the role of the identity of the sender of the request is a patient. In some aspects, censored results may be generated using the method 300 in other circumstances (e.g., when the results will be shared with another party that is not authorized to view uncensored results, for example due to data privacy laws). The rules engine may include additional rules governing these scenarios, in some aspects.

[0062] In some aspects, generating the modified sequence data by processing the sequence data using the configured rules engine may include generating annotated results when the role of the identity of the sender of the request is a clinician. Specifically, the annotated results may include a

binary indication (e.g., call/no call) and/or additional indicia of the quality control check on the results. For example, the additional indicia may include or relate to the predicted standard deviation values, the classifications of the simulated expression values (e.g., count information), and/or the confidence scores. Whereas this information may be censored or not displayed for a patient, the physician or another viewer may have more insight into the underlying quality control techniques, and the information may prove useful for interpretation. In some aspects, the annotations may include an indication of a patient therapy, including one or more alternative therapy as discussed below.

[0063] In some aspects, the method 300 may be performed automatically as part of a bioinformatics pipeline, wherein the role of the sender of the request is a programmatic script; and wherein generating the modified sequence data by processing the sequence data may include programmatically generating a structured output corresponding to the modified sequence data, wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data. For example the structured output may be an XML output, a JSON output, a CSV output or another structured data format. The method 300 may include serving the structured output via an authenticated API (e.g., the API 112) to a requesting client (e.g., the client computing device 104) via an electronic network (e.g., the network 106) in some aspects.

[0064] In some aspects, the number of classifications is two or more. For example, the method 300 may include classifications with respect to two subtypes, a subtype A and a subtype B, as discussed herein. In some aspects, the classifications may be multi-stage classifications.

[0065] In some aspects, the method 300 may include determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be an appropriate match for the patient corresponding to the sample. The alternative therapy may include at least one of: (i) a first-line therapy or (ii) a second-line therapy. The alternative therapy may include at least one of: (i) a chemotherapeutic drug therapy, (ii) a radiation therapy, (iii) a surgery therapy, (iv) a therapy in a different class, (v) a molecular inhibitor therapy, (vi) an antibody therapy, (vii) a recombinant nucleic acid therapy, (viii) an engineered immune cell therapy, (ix) a checkpoint inhibitor therapy, (x) a cytokine therapy, (xi) a cancer treatment vaccine therapy, (xii) a CAR-cell therapy; or (xiii) an oncolytic virus therapy.

[0066] Additional exemplary therapeutic agents that can be used with the methods described herein are provided in Table 2. Additional alternative therapies are discussed in U.S. Pat. No. 11,415,571, entitled “Large scale organoid analysis,” filed on Dec. 7, 2020, and incorporated herein by reference in its entirety for all purposes.

TABLE 2

Agent	Target	Pathway	Formula
Veliparib (ABT-888)	PARP	DNA Damage	C ₁₃ H ₁₆ N ₄ O
Selumetinib (AZD6244)	MEK	MAPK	C ₁₇ H ₁₅ BrClFN ₄ O ₃
PD184352 (CI-1040)	MEK	MAPK	C ₁₇ H ₁₄ ClF ₂ IN ₂ O ₂
PD0325901	MEK	MAPK	C ₁₆ H ₁₄ F ₃ IN ₂ O ₄
Tozasertib (VX-680, MK-0457)	Aurora Kinase	Cell Cycle	C ₂₃ H ₂₈ N ₈ OS
Y-27632 2HCl	Autophagy, ROCK	Cell Cycle	C ₁₄ H ₂₃ C ₁₂ N ₃ O
Olaparib (AZD2281, Ku-0059436)	PARP	DNA Damage	C ₂₄ H ₂₃ FN ₄ O ₃

TABLE 2-continued

Agent	Target	Pathway	Formula
SL-327	MEK	MAPK	C ₁₆ H ₁₂ F ₃ N ₃ S
SB431542	TGF-beta/Smad	TGF-beta/Smad	C ₂₂ H ₁₆ N ₄ O ₃
MK-2206 2HCl	Akt	PI3K/Akt/mTOR	C ₂₅ H ₂₃ C ₁₂ N ₅ O
Refametinib (RDEA119, Bay 86-9766)	MEK	MAPK	C ₁₉ H ₂₀ F ₃ IN ₂ O ₅ S
KU-55933 (ATM Kinase Inhibitor)	ATM/ATR	DNA Damage	C ₂₁ H ₁₇ NO ₃ S ₂
GSK1904529A	IGF-1R	Protein Tyrosine Kinase	C ₄₄ H ₄₇ F ₂ N ₉ O ₅ S
PF-04217903	c-Met	Protein Tyrosine Kinase	C ₁₉ H ₁₆ N ₈ O
U0126-EtOH	MEK	MAPK	C ₂₀ H ₂₂ N ₆ OS ₂
BI 2536	PLK	Cell Cycle	C ₂₈ H ₃₉ N ₇ O ₃
JNJ-38877605	c-Met	Protein Tyrosine Kinase	C ₁₉ H ₁₃ F ₂ N ₇
Odanacatib (MK-0822)	Cysteine Protease	Proteases	C ₂₅ H ₂₇ F ₄ N ₃ O ₃ S
Alisertib (MLN8237)	Aurora Kinase	Cell Cycle	C ₂₇ H ₂₀ ClFN ₄ O ₄
Barasertib (AZD1152-HQPA)	Aurora Kinase	Cell Cycle	C ₂₆ H ₃₀ FN ₇ O ₃
CP-724714	EGFR, HER2	Protein Tyrosine Kinase	C ₂₇ H ₂₇ N ₅ O ₃
TGX-221	PI3K	PI3K/Akt/mTOR	C ₂₁ H ₂₄ N ₄ O ₂
WZ4002	EGFR	Protein Tyrosine Kinase	C ₂₅ H ₂₇ ClN ₆ O ₃
BIBR 1532	Telomerase	DNA Damage	C ₂₁ H ₁₇ NO ₃
Anastrozole	Aromatase	Endocrinology & Hormones	C ₁₇ H ₁₉ N ₅
Aprepitant	Substance P	Others	C ₂₃ H ₂₁ F ₇ N ₄ O ₃
TAK-700 (Orteronel)	P450 (e.g. CYP17)	Metabolism	C ₁₈ H ₁₇ N ₃ O ₂
PFI-1 (PF-6405761)	Epigenetic Reader Domain	Epigenetics	C ₁₆ H ₁₇ N ₃ O ₄ S
KU-0063794	mTOR	PI3K/Akt/mTOR	C ₂₅ H ₃₁ N ₅ O ₄
CHIR-99021 (CT99021)	GSK-3	PI3K/Akt/mTOR	C ₂₂ H ₁₈ Cl ₂ N ₈
WYE-354	mTOR	PI3K/Akt/mTOR	C ₂₄ H ₂₉ N ₇ O ₅
TG100-115	PI3K	PI3K/Akt/mTOR	C ₁₈ H ₁₄ N ₆ O ₂
Aurora A Inhibitor I	Aurora Kinase	Cell Cycle	C ₃₁ H ₃₁ ClFN ₇ O ₂
Ispinesib (SB-715992)	Kinesin	Cytoskeletal Signaling	C ₃₀ H ₃₃ ClN ₄ O ₂
Zibotentan (ZD4054)	Endothelin Receptor	GPCR & G Protein	C ₁₉ H ₁₆ N ₆ O ₄ S
Safinamide Mesylate	MAO	Metabolism	C ₁₈ H ₂₃ FN ₂ O ₅ S
GSK429286A	ROCK	Cell Cycle	C ₂₁ H ₁₆ F ₄ N ₄ O ₂
Pimasertib (AS-703026)	MEK	MAPK	C ₁₅ H ₁₅ FIN ₃ O ₃
Tadalafil	PDE	Metabolism	C ₂₂ H ₁₉ N ₃ O ₄
Adavosertib (MK-1775)	Wee1	Cell Cycle	C ₂₇ H ₃₂ N ₈ O ₂
CP-673451	PDGFR	Protein Tyrosine Kinase	C ₂₄ H ₂₇ N ₅ O ₂
Selisistat (EX 527)	Sirtuin	Epigenetics	C ₁₃ H ₁₃ ClN ₂ O
Dapagliflozin	SGLT	GPCR & G Protein	C ₂₁ H ₂₅ ClO ₆
Nebivolol HCl	Adrenergic Receptor	Neuronal Signaling	C ₂₂ H ₂₆ ClF ₂ N ₄ O ₄
Pimobendan	PDE	Metabolism	C ₁₉ H ₁₈ N ₄ O ₂
AZD8055	mTOR	PI3K/Akt/mTOR	C ₂₅ H ₃₁ N ₅ O ₄
KU-60019	ATM/ATR	DNA Damage	C ₃₀ H ₃₃ N ₃ O ₅ S
Tie2 kinase inhibitor	Tie-2	ProteinTyrosine Kinase	C ₂₆ H ₂₁ N ₃ O ₂ S
Apixaban	Factor Xa	Metabolism	C ₂₅ H ₂₅ N ₅ O ₄
Raltegravir (MK-0518)	Integrase	Microbiology	C ₂₀ H ₂₁ FN ₆ O ₅
PCI-34051	HDAC	Epigenetics	C ₁₇ H ₁₆ N ₂ O ₃
Ambrisentan	Endothelin Receptor	GPCR & G Protein	C ₂₂ H ₂₂ N ₂ O ₄
SB743921 HCl	Kinesin	Cytoskeletal Signaling	C ₃₁ H ₃₄ Cl ₂ N ₂ O ₃
AST-1306	EGFR	Protein Tyrosine Kinase	C ₃₁ H ₂₆ ClFN ₄ O ₅ S
Sapitinib (AZD8931)	EGFR, HER2	Protein Tyrosine Kinase	C ₂₃ H ₂₅ ClFN ₅ O ₃
GSK461364	PLK	Cell Cycle	C ₂₇ H ₂₈ F ₃ N ₅ O ₂ S
Mubritinib (TAK 165)	HER2	Protein Tyrosine Kinase	C ₂₅ H ₂₃ F ₃ N ₄ O ₂
UK 383367	Procollagen C Proteinase	Metabolism	C ₁₅ H ₂₄ N ₄ O ₄

TABLE 2-continued

Agent	Target	Pathway	Formula
Cryptotanshinone	STAT	JAK/STAT	C19H20O3
Icariin	PDE	Metabolism	C33H40O15
OSI-027	mTOR	PI3K/Akt/mTOR	C21H22N6O3
Rabusertib (LY2603618)	Chk	Cell Cycle	C18H22BrN5O3
URB597	FAAH	Metabolism	C20H22N2O3
A66	PI3K	PI3K/Akt/mTOR	C17H23N5O2S2
ICG-001	Wnt/beta-catenin	Stem Cells & Wnt	C33H32N4O4
PF-3845	FAAH	Metabolism	C24H23F3N4O2
Trametinib (GSK1120212)	MEK	MAPK	C26H23FIN5O4
Ibrutinib (PCI-32765)	BTK	Angiogenesis	C25H24N6O2
CHIR-124	Chk	Cell Cycle	C23H22CIN5O
Mardepodect (PF- 2545920)	PDE	Metabolism	C25H20N4O
WAY-600	mTOR	PI3K/Akt/mTOR	C28H30N8O
Nepicastat (SYN-117) HCl	Hydroxylase	Metabolism	C14H16CIF2N3S
RS-127445	5-HT Receptor	Neuronal Signaling	C17H16FN3
CP-91149	Phosphorylase	Metabolism	C21H22CIN3O3
SB415286	GSK-3	PI3K/Akt/mTOR	C16H10CIN3O5
GSK1070916	Aurora Kinase	Cell Cycle	C30H33N7O
Niraparib (MK-4827)	PARP	DNA Damage	C19H20N4O
CHIR-98014	GSK-3	PI3K/Akt/mTOR	C20H17Cl2N9O2
AMG-458	c-Met	Protein Tyrosine Kinase	C30H29N5O5
Tivantinib (ARQ 197)	c-Met	Protein Tyrosine Kinase	C23H19N3O2
Canagliflozin	SGLT	GPCR & G Protein	C24H25FO5S
NVP-BVU972	c-Met	Protein Tyrosine Kinase	C20H16N6
MK-5108 (VX-689)	Aurora Kinase	Cell Cycle	C ₂₂ H ₂₁ ClFN ₃ O ₃ S
SB705498	TRPV	Others	C17H16BrF3N4O
Vistusertib (AZD2014)	mTOR	PI3K/Akt/mTOR	C25H30N6O3
A-803467	Sodium Channel	Transmembrane Transporters	C19H16CINO4
Sirtinol	Sirtuin	Epigenetics	C26H22N2O2
Ipatasertib (GDC-0068)	Akt	PI3K/Akt/mTOR	C24H32CIN5O2
Sapanisertib (INK 128, MLN0128)	mTOR	PI3K/Akt/mTOR	C15H15N7O
Tyrphostin AG 879	HER2	Protein Tyrosine Kinase	C18H24N2OS
JNJ-1661010	FAAH	Metabolism	C19H19N5OS
CTEP (RO4956371)	GluR	Neuronal Signaling	C19H13CIF3N3O
Alogliptin (SYK-322) benzoate	DPP-4	Proteases	C18H21N5O2
T0070907	PPAR	DNA Damage	C12H8CIN3O3
GW441756	Trk receptor	Protein Tyrosine Kinase	C17H13N3O
SB742457	5-HT Receptor	Neuronal Signaling	C19H19N3O2S
ZM 323881 HCl	VEGFR	Protein Tyrosine Kinase	C22H19CIFN3O2
GNF-2	Bcr-Abl	Angiogenesis	C18H13F3N4O2
Lumiracoxib	COX	Neuronal Signaling	C15H13CIFNO2
JNJ-7777120	Histamine Receptor	Neuronal Signaling	C14H16CIN3O
IOX2	HIF	Angiogenesis	C19H16N2O5
PF-4981517	P450 (e.g. CYP17)	Metabolism	C26H32N8
CHIR-99021 (CT99021) HCl	GSK-3	PI3K/Akt/mTOR	C22H19Cl3N8
Rivaroxaban	Factor Xa	Metabolism	C19H18CIN3O5S
Linagliptin	DPP-4	Proteases	C25H28N8O2
Azilsartan Medoxomil	RAAS	Endocrinology & Hormones	C30H24N4O8
Sulfaphenazole	P450 (e.g. CYP17)	Metabolism	C15H14N4O2S
Sitagliptin phosphate monohydrate	DPP-4	Proteases	C16H20F6N5O6P

TABLE 2-continued

Agent	Target	Pathway	Formula
Avanafil	PDE	Metabolism	C23H26ClN7O3
Eprosartan Mesylate	RAAS	Endocrinology & Hormones	C24H28N2O7S2
Carprofen	COX	Neuronal Signaling	C15H12ClNO2
Saxagliptin hydrate	DPP-4	Proteases	C18H27N3O3
Daminozide	Histone Demethylase	Epigenetics	C6H12N2O3
Bedaquiline fumarate	Anti-infection	Microbiology	C36H35BrN2O6
JZL184	Lipase	Metabolism	C27H24N2O9
SC-514	IκB/IKK	NF-κB	C9H8N2OS2
(R)-Nepicastat HCl	Hydroxylase	Metabolism	C14H16ClF2N3S
Asunaprevir	HCV Protease	Proteases	C35H46ClN5O9S
Trelagliptin succinate	DPP-4	Proteases	C22H26FN5O6
Dabrafenib Mesylate	Raf	MAPK	C ₂₄ H ₂₄ F ₃ N ₅ O ₅ S ₃
Argatroban	Thrombin	Others	C23H38N6O6S
Monohydrate			
Sitagliptin	DPP-4	Proteases	C16H15F6N5O
Raltegravir potassium	Integrase, HIV Protease	Microbiology	C20H20FKN6O5
Alogliptin	DPP-4	Proteases	C18H21N5O2
Dasabuvir (ABT-333)	HCV Protease	Proteases	C26H27N3O5S
Ertugliflozin	SGLT2	Ion-Channel	C22H25ClO7
Rolapitant	NK1-receptor	GPCR	C25H26F6N2O2
Dapagliflozin	SGLT	GPCR & G	C24H35ClO9
propanediol monohydrate		Protein	
Bedaquiline	tuberculosis	Immunology	C32H31BrN2O2
Fruquintinib	VEGFRs	VEGFR	C21H19N3O5
JNJ0966	Others	Others	C16H16N4O2S2
acalisib (GS-9820)	PI3K	PI3K/Akt/mTOR	C21H16FN7O
BRL-50481	PDE	Metabolism	C9H12N2O4S
Canagliflozin hemihydrate	SGLT	GPCR & G	C48H52F2O11S2
		Protein	
JANEX-1	JAK	JAK/STAT	C16H15N3O3
Anagliptin	DPP-4	Proteases	C19H25N7O2
GSK 5959	Epigenetic Reader Do	Epigenetics	C22H26N4O3
Pitolisant hydrochloride	Histamine Receptor	Neuronal	C17H27Cl2NO
		Signaling	
K 858	Kinesin	Cytoskeletal Signaling	C13H15N3O2S
BAY-61-3606	Syk	Angiogenesis	C20H20Cl2N6O3
Stattic	STAT	JAK/STAT	C8H5NO4S
GSK2656157	PERK	Apoptosis	C23H21FN6O
XL388	mTOR	PI3K/Akt/mTOR	C23H22FN3O4S
LY2090314	GSK-3	PI3K/Akt/mTOR	C28H25FN6O3
MK-8745	Aurora Kinase	Cell Cycle	C20H19ClFN5OS
Tepotinib (EMD 1214063)	c-Met	Protein Tyrosine	C29H28N6O2
		Kinase	
SGC 0946	Histone Methyltransferase	Epigenetics	C28H40BrN7O4
GSK2334470	PDK	PI3K/Akt/mTOR	C25H34N8O
IPA-3	PAR	Cytoskeletal Signaling	C20H14O2S2
VE-822	ATM/ATR	PI3K/Akt/mTOR	C24H25N5O3S
(+)-JQ1	Epigenetic Reader Domain	Epigenetics	C23H25ClN4O2S
PYR-41	E1 Activating	Ubiquitin	C17H13N3O7
TCID	DUB	Ubiquitin	C9H2Cl4O2
DMH1	TGF-beta/Smad	TGF-beta/Smad	C24H20N4O
ML347	TGF-beta/Smad, ALK	TGF-beta/Smad	C22H16N4O
UNC1999	Histone Methyltransferase	Epigenetics	C33H43N7O2
SSR128129E	FGFR	Angiogenesis	C18H15N2NaO4
Spebrutinib (CC-292, AVL-292)	BTK	Angiogenesis	C22H22FN5O3
SKI II	S1P Receptor	GPCR & G	C15H11ClN2OS
		Protein	
PF-543	S1P Receptor	GPCR & G	C27H31NO4S
		Protein	
CID755673	Serine/threonine kinase, CaMK	Apoptosis	C12H11NO3

TABLE 2-continued

Agent	Target	Pathway	Formula
1-Azakenpauillone	GSK-3	PI3K/Akt/mTOR	C15H10BrN3O
CNX-2006	EGFR	Protein Tyrosine Kinase	C26H27F4N7O2
Bisindolylmaleimide I (GF109203X)	PKC	TGF-beta/Smad	C25H24N4O2
Thiamet G	Others	Others	C9H16N2O4S
Alvelestat (AZD9668)	Serine Protease	Proteases	C24H20F3N5O4S
RGFP966	HDAC	Epigenetics	C21H19FN4O
UNC0642	Histone Methyltransferase	Epigenetics	C29H44F2N6O2
NVP-TNKS656	PARP	DNA Damage	C27H34N4O5
AGI-6780	Dehydrogenase	Metabolism	C ₂₁ H ₁₈ F ₃ N ₃ O ₃ S ₂
Ro3280	PLK	Cell Cycle	C27H35F2N7O3
NMS-P937 (NMS1286937)	PLK	Cell Cycle	C24H27F3N8O3
CNX-774	BTK	Angiogenesis	C26H22FN7O3
AZD1981	GPR	Endocrinology & Hormones	C19H17ClN2O3S
SRPIN340	Others	Others	C18H18F3N3O
4μ8C	Others	Others	C11H8O4
NMS-E973	HSP (e.g. HSP90)	Cytoskeletal Signaling	C22H22N4O7
PFI-2 HCl	Histone Methyltransferase	Epigenetics	C23H25F4N3O3S
GSK2606414	PERK	Apoptosis	C24H20F3N5O
IPI-3063	PI3K	PI3K/Akt/mTOR	C25H25N7O2
Atglistatin	Lipase	Metabolism	C17H21N3O
CGP 57380	MNK	MAPK	C11H9FN6
SB-3CT	MMP	Proteases	C15H14O3S2
AR-A014418	GSK-3	PI3K/Akt/mTOR	C12H12N4O4S
NH125	CaMK	Neuronal Signaling	C27H45IN2
XEN445	Lipase	Metabolism	C18H17F3N2O3R
LDC000067	CDK	Cell Cycle	C18H18N4O3S
PI-1840	Proteasome	Proteases	C22H26N4O3
FTI 277 HCl	Transferase	Metabolism	C ₂₂ H ₃₀ C ₇ N ₃ O ₃ S ₂
Nexturastat A	HDAC	DNA Damage	C19H23N3O3
ESI-09	cAMP	GPCR & G Protein	C16H15ClN4O2
HJC0350	cAMP	GPCR & G Protein	C15H19NO2S
HO-3867	STAT	JAK/STAT	C28H30F2N2O2
JNK Inhibitor IX	JNK	MAPK	C20H16N2OS
Trelagliptin	DPP-4	Proteases	C18H20FN5O2
XMD8-92	ERK	MAPK	C26H30N6O3
A-366	Histone Methyltransferase	Epigenetics	C19H27N3O2
GSK-LSD1 2HCl	Histone Demethylase	Epigenetics	C14H22Cl2N2
LLY-507	Histone Methyltransferase	Epigenetics	C36H42N6O
Santacruzamate A (CAY10683)	HDAC	DNA Damage	C15H22N2O3
CAY10603	HDAC	DNA Damage	C22H30N4O6
GSK1324726A (I-BET726)	Epigenetic Reader	Epigenetics	C25H23ClN2O3
SD-208	Domain	TGF-beta/Smad	C17H10ClFN6
TH588	MTH1	DNA Damage	C13H12Cl2N4
SB225002	CXCR	GPCR & G Protein	C13H10BrN3O4
CPI-360	Histone Methyltransferase	Epigenetics	C25H31N3O4
Picropodophyllin (PPP)	IGF-1R	Protein Tyrosine Kinase	C22H22O8
Savolitinib(AZD6094, HMPL-504)	c-Met	Protein Tyrosine Kinase	C17H15N9
SP2509	Histone Demethylase	Epigenetics	C19H20ClN3O5S
VX-11e	ERK	MAPK	C ₂₄ H ₂₀ C ₁₂ FN ₅ O ₂
SBE 13 HCl	PLK	Cell Cycle	C24H28Cl2N2O4
BLZ945	CSF-1R	Protein Tyrosine Kinase	C20H22N4O3S
LFM-A13	BTK	Angiogenesis	C11H8Br2N2O2
EPZ015666(GSK3235025)	Histone Methyltransferase	Epigenetics	C20H25N5O3

TABLE 2-continued

Agent	Target	Pathway	Formula
VER155008	HSP (e.g. HSP90)	Cytoskeletal Signaling	C25H23Cl2N7O4
BPES	Glutaminase	Proteases	C24H24N6O2S3
AZ6102	PPAR	DNA Damage	C25H28N6O
Erlotinib	EGFR	Protein Tyrosine Kinase	C22H23N3O4
ORY-1001 (RG-6016) 2HCl	Histone Demethylase	Epigenetics	C15H24Cl2N2
EPZ020411 2HCl	Histone Methyltransferase	Epigenetics	C25H40Cl2N4O3
I-BRD9	Epigenetic Reader Domain	Epigenetics	C ₂₂ H ₂₂ F ₃ N ₃ O ₃ S ₂
SirReal2	Sirtuin	Epigenetics	C22H20N4OS2
BDA-366	Bcl-2	Apoptosis	C24H29N3O4
NVP-CGM097	Mdm2	Apoptosis	C38H47ClN4O4
CC-223	mTOR	PI3K/Akt/mTOR	C21H27N5O3
PFI-4	Epigenetic Reader Domain	Epigenetics	C21H24N4O3
BIO-acetoxime	GSK-3	PI3K/Akt/mTOR	C18H12BrN3O3
GSK2292767	PI3K	PI3K/Akt/mTOR	C24H28N6O5S
SIS3 HCl	TGF-beta/Smad	TGF-beta/Smad	C28H28ClN3O3
Larotrectinib (LOXO-101) sulfate	Trk receptor	Protein Tyrosine Kinase	C21H24F2N6O6S
PLX7904	Raf	MAPK	C24H22F2N6O3S
VPS34-IN1	PI3K	PI3K/Akt/mTOR	C21H24ClN7O
A-196	Histone Methyltransferase	Epigenetics	C18H16Cl2N4
LDC4297 (LDC044297)	CDK	Cell Cycle	C23H28N8O
SMI-4a	Pim	JAK/STAT	C11H6F3NO2S
Empagliflozin (BI 10773)	SGLT	GPCR & G	C23H27ClO7
TCS 359	FLT3	Protein Angiogenesis	C18H20N2O4S
NSC 23766	Rho	Cell Cycle	C24H38Cl3N7
GDC-0349	mTOR	PI3K/Akt/mTOR	C24H32N6O3
Cobimetinib (GDC-0973, RG7420)	MEK	MAPK	C21H21F3IN3O2
GW2580	CSF-1R	Protein Tyrosine Kinase	C20H22N4O3
BMS-345541	IκB/IKK	NF-κB	C14H17N5
Dynasore	Dynamin	Cytoskeletal Signaling	C18H14N2O4
Venetoclax (ABT-199, GDC-0199)	Bcl-2	Apoptosis	C45H50ClN7O7S
ICI-118551 Hydrochloride	Adrenergic Receptor	GPCR & G	C17H28ClNO2
AMG 337	c-Met	Protein Tyrosine Kinase	C23H22FN7O3
PF-CBP1 HCl	Epigenetic Reader Domain	Epigenetics	C29H37ClN4O3
CPI-637	Epigenetic Reader Domain	Epigenetics	C22H22N6O
BI-78D3	JNK	MAPK	C13H9N5O5S2
SB366791	TRPV	Transmembrane Transporters	C16H14ClNO2
Thiomristoyl	Sirtuin	DNA Damage	C34H51N3O3S
CCT245737	Chk	Cell Cycle	C16H16F3N7O
GSK6853	Epigenetic Reader Domain	Epigenetics	C22H27N5O3
SHP099 dihydrochloride	phosphatase	Others	C16H21Cl4N5
Selonsertib (GS-4997)	ASK	Apoptosis	C24H24FN7O
KYA1797K	Wnt/beta-catenin	Stem Cells & Wnt	C17H11KN2O6S2
IPI-549	PI3K	PI3K/Akt/mTOR	C30H24N8O2
SGC2085	Histone Methyltransferase	Epigenetics	C19H24N2O2
Irbinitinib (ARRY-380, ONT-380)	HER2	Protein Tyrosine Kinase	C26H24N8O2

TABLE 2-continued

Agent	Target	Pathway	Formula
NMS-P118	PARP	DNA Damage	C ₂₀ H ₂₄ F ₃ N ₃ O ₂
BAY-876	GLUT	Metabolism	C ₂₄ H ₁₆ F ₄ N ₆ O ₂
VPS34 inhibitor 1 (Compound 19, PIK-III analogue)	PI3K	PI3K/Akt/mTOR	C ₂₁ H ₂₅ N ₇ O
UK-371804 HCl	Serine Protease	Proteases	C ₁₄ H ₁₇ Cl ₂ N ₅ O ₄ S
GSK'872 (GSK2399872A)	Serine/threonine kinase	Apoptosis	C ₁₉ H ₁₇ N ₃ O ₂ S ₂
LLY-283	Histone Methyltransferase	Epigenetics	C ₁₇ H ₁₈ N ₄ O ₄
GSK180736A (GSK180736)	ROCK	Cell Cycle	C ₁₉ H ₁₆ FN ₅ O ₂
PD-166866 (PD166866)	FGFR	Angiogenesis	C ₂₀ H ₂₄ N ₆ O ₃
BLU-554 (BLU554)	FGFR	Angiogenesis	C ₂₄ H ₂₄ Cl ₂ N ₄ O ₄
LY3214996	ERK	MAPK	C ₂₂ H ₂₇ N ₇ O ₂ S
PF-06651600	JAK	JAK/STAT	C ₁₅ H ₁₉ N ₅ O
FM-381	JAK	JAK/STAT	C ₂₄ H ₂₄ N ₆ O ₂
AZ31	ATM/ATR	DNA Damage	C ₂₄ H ₂₈ N ₄ O ₃
Tofogliflozin(CSG 452)	SGLT	GPCR & G Protein	C ₂₂ H ₂₈ O ₇
Omarigliptin (MK-3102)	DPP-4	Proteases	C ₁₇ H ₂₀ F ₂ N ₄ O ₃ S
Serabelisib (INK- 1117, MLN-1117, TAK- 117)	PI3K	PI3K/Akt/mTOR	C ₁₉ H ₁₇ N ₅ O ₃
FX1	Bcl-6	Apoptosis	C ₁₄ H ₉ CIN ₂ O ₄ S ₂
Pamiparib (BGB-290)	PARP	DNA Damage	C ₁₆ H ₁₅ FN ₄ O
NCT-503	Dehydrogenase	Metabolism	C ₂₀ H ₂₃ F ₃ N ₄ S
Chk2 Inhibitor II (BML- 277)	Chk	Cell Cycle	C ₂₀ H ₁₄ CIN ₃ O ₂
Ipragliflozin (ASP1941)	SGLT	GPCR & G Protein	C ₂₁ H ₂₁ FO ₅ S
Nec-1s (7-Cl—O—Nec1)	TNF-alpha	Apoptosis	C ₁₃ H ₁₂ CIN ₃ O ₂
GSK'963	NF-kB, TNF-alpha	NF-kB	C ₁₄ H ₁₈ N ₂ O
GNF-6231	Wnt/beta-catenin	Stem Cells & Wnt	C ₂₄ H ₂₅ FN ₆ O ₂
Skp2 inhibitor C1 (SKPin C1)	CDK	Cell Cycle	C ₁₈ H ₁₃ BrN ₂ O ₄ S ₂
PF-06840003	IDO	Metabolism	C ₁₂ H ₉ FN ₂ O ₂
GI254023X	Immunology & Inflammation related	Immunology & Inflammation	C ₂₁ H ₃₃ N ₃ O ₄
BAY 1895344 (BAY- 1895344)	ATM/ATR	DNA Damage	C ₂₀ H ₂₂ CIN ₇ O
Tenalisib (RP6530)	PI3K	PI3K/Akt/mTOR	C ₂₃ H ₁₈ FN ₅ O ₂
H3B-6527	FGFR	Protein Tyrosine Kinase	C ₂₉ H ₃₄ Cl ₂ N ₈ O ₄
Cu-CPT22	TLR	Immunology & Inflammation	C ₁₉ H ₂₂ O ₇
AZD1390	ATM/ATR	PI3K/Akt/mTOR	C ₂₇ H ₃₂ FN ₅ O ₂
Atuveciclib (BAY- 1143572)	CDK	Cell Cycle	C ₁₈ H ₁₈ FN ₅ O ₂ S
LXH254	Raf	MAPK	C ₂₅ H ₂₅ F ₃ N ₄ O ₄
WM-1119	Histone Acetyltransf	Epigenetics	C ₁₈ H ₁₃ F ₂ N ₃ O ₃ S
Evobrutinib	BTk	Protein Tyrosine Kinase	C ₂₅ H ₂₇ N ₅ O ₂
LIT-927	CXCR	Immunology & Inflammation	C ₁₇ H ₁₃ CIN ₂ O ₃
4-Hydroxyquinazoline	Others	antiplatelet	C ₈ H ₆ N ₂ O
Valbenazine tosylate	VMAT2	Others	C ₃₈ H ₅₄ N ₂ O ₁₀ S ₂
SAR125844	c-Met	Protein Tyrosine Kinase	C ₂₅ H ₂₃ FN ₈ O ₂ S ₂
dBET1	Epigenetic Reader Do	Epigenetics	C ₃₈ H ₃₇ CIN ₈ O ₇ S
GSK'547	TNF-alpha	Apoptosis	C ₂₀ H ₁₈ F ₂ N ₆ O
Palbociclib (PD- 0332991)	CDK	Cell Cycle	C ₂₄ H ₃₀ CIN ₇ O ₂
HCl			
Palbociclib (PD0332991)	CDK	Cell Cycle	C ₂₆ H ₃₅ N ₇ O ₆ S
Isethionate			
LDN-193189 2HCl	TGF-beta/Smad	TGF-beta/Smad	C ₂₅ H ₂₃ CIN ₆
MCC950(CP-456773)	Immunology & Inflammation related	Immunology & Inflammation	C ₂₀ H ₂₃ N ₂ NaO ₅ S
bpV (HOpic)	PTEN	Others	C ₆ H ₄ K ₂ N ₈ O ₈ V
Erlotinib HCl (OSI-744)	Autophagy, EGFR	Protein Tyrosine Kinase	C ₂₂ H ₂₄ CIN ₃ O ₄

TABLE 2-continued

Agent	Target	Pathway	Formula
SGX-523	c-Met	Protein Tyrosine Kinase	C18H13N7S
(-)-Huperzine A (HupA)	GluR, AChR	Neuronal Signaling	C15H18N2O
GSK256066	PDE	Metabolism	C27H26N4O5S
CCT137690	Aurora Kinase	Cell Cycle	C26H31BrN8O
Capmatinib (INCB28060)	c-Met	Protein Tyrosine Kinase	C23H17FN6O
EPZ005687	Histone Methyltransferase	Epigenetics	C32H37N5O3
GSK126	Histone Methyltransferase	Epigenetics	C31H38N6O2
Tazemetostat (EPZ-6438)	Histone Methyltransferase	Epigenetics	C34H44N4O4
ISRIB (trans-isomer)	PERK	Apoptosis	C22H24Cl2N2O4
A-1210477	Bcl-2	Apoptosis	C46H55N7O7S
Otenabant (CP-945598) HCl	Cannabinoid Receptor	GPCR & G	C25H26Cl3N7O
FGF401	FGFR	Protein Tyrosine Kinase	C ₂₅ H ₃₀ N ₈ O ₄
Lazertinib (YH25448, GNS-1480)	EGFR	Protein Tyrosine Kinase	C ₃₀ H ₃₄ N ₈ O ₃

[0067] Aspects of the techniques described in the present disclosure may include any of the following aspects, either alone or in combination:

[0068] 1. A computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, comprising: receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality of samples: computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; computing, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample; and storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0069] 2. The computer-implemented method of aspect 1, further comprising: generating, via one or more processors, a static digital report including at least one classification result.

[0070] 3. The computer-implemented method of any of aspects 1-2, wherein generating the static digital report includes censoring a no-call classification result.

[0071] 4. The computer-implemented method of any of aspects 1-3, further comprising: receiving, via an application programming interface, a request for RNA sequence data; processing, via one or more processors, the request using a rules engine to determine an identity of a sender of the

request; determining, via one or more processors, a role of the identity of the sender of the request; configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request.

[0072] 5. The computer-implemented method of any of aspects 1-4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating censored results when the role of the identity of the sender of the request is a patient.

[0073] 6. The computer-implemented method of any of aspects 1-4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating annotated results when the role of the identity of the sender of the request is a clinician.

[0074] 7. The computer-implemented method of any of aspects 1-4, wherein the method is performed automatically as part of a bioinformatics pipeline; wherein the role of the sender of the request is a programmatic script; and wherein generating the modified sequence data by processing the sequence data includes programmatically generating a structured output corresponding to the modified sequence data, wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data.

[0075] 8. The computer-implemented method of any of aspects 1-7, wherein the number of classifications is two or more.

[0076] 9. The computer-implemented method of any of aspects 1-8, further comprising: determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

[0077] 10. The computer-implemented method of any of aspects 1-9, wherein the alternative therapy includes at least one of: (i) a first-line therapy or (ii) a second-line therapy.

[0078] 11. The computer-implemented method of any of aspects 1-10, wherein the alternative therapy includes at least one of: (i) a chemotherapeutic drug therapy, (ii) a radiation therapy, (iii) a surgery therapy, (iv) a therapy in a different class, (v) a molecular inhibitor therapy, (vi) an antibody therapy, (vii) a recombinant nucleic acid therapy, (viii) an engineered immune cell therapy, (ix) a checkpoint inhibitor therapy, (x) a cytokine therapy, (xi) a cancer treatment vaccine therapy, (xii) a CAR-cell therapy; or (xiii) an oncolytic virus therapy.

[0079] 12. The computer-implemented method of any of aspects 1-11, wherein the machine learning model is at least one of a Bayesian model, a supervised machine learning model or an unsupervised machine learning model.

[0080] 13. A computing system, comprising: one or more processors; and

[0081] one or more memories, having stored thereon instructions that when executed, cause the computing system to: receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via the one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0082] 14. The computing system of aspect 13, the memory having stored thereon instructions that when executed, cause the computing system to: generate, via one or more processors, a static digital report including at least one classification result.

[0083] 15. The computing system of any of aspects 13-14, the memory having stored thereon instructions that when executed, cause the computing system to: receive, via an application programming interface, a request for RNA sequence data; process, via one or more processors, the request using a rules engine to determine an identity of a sender of the request; determine, via one or more processors, a role of the identity of the sender of the request; configure, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generate, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmit, via electronic network, the modified sequence data to a computing device of the sender of the request.

[0084] 16. The computing system of any of aspects 13-15, the memory having stored thereon instructions that when

executed, cause the computing system to: generate censored results when a role of an identity of a sender of a request is a patient.

[0085] 17. The computing system of any of aspects 13-16, the memory having stored thereon instructions that when executed, cause the computing system to: generate annotated results when a role of an identity of a sender of a request is a clinician.

[0086] 18. The computing system of aspects 13-17, wherein the number of classifications is two or more.

[0087] 19. The computing system of aspects 13-18, the memory having stored thereon instructions that when executed, cause the computing system to: determine, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

[0088] 20. A computer-readable media having stored thereon a set of non-transitory computer-readable instructions that, when executed, cause a computer to: receive, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classify, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

Additional Considerations

[0089] The computer-readable media may include executable computer-readable code stored thereon for programming a computer (e.g., comprising a processor(s) and GPU(s)) to the techniques herein. Examples of such computer-readable storage media include a hard disk, a CD-ROM, digital versatile disks (DVDs), an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. More generally, the processing units of the computing device **1300** may represent a CPU-type processing unit, a GPU-type processing unit, a TPU-type processing unit, a field-programmable gate array (FPGA), another class of digital signal processor (DSP), or other hardware logic components that can be driven by a CPU.

[0090] A system for performing the methods described herein may include a computing device, and more particularly may be implemented on one or more processing units, for example, Central Processing Units (CPUs), and/or on one or more or Graphical Processing Units (GPUs), including clusters of CPUs and/or GPUs. Features and functions described may be stored on and implemented from one or

more non-transitory computer-readable media of the computing device. The computer-readable media may include, for example, an operating system and software modules, or “engines,” that implement the methods described herein. Those engines may be stored as sets of non-transitory computer-executable instructions. The computing device may be a distributed computing system, such as an Amazon Web Services, Google Cloud Platform Microsoft Azure, or other public, private and/or hybrid cloud computing solution.

[0091] The computing device includes a network interface communicatively coupled to network, for communicating to and/or from a portable personal computer, smart phone, electronic document, tablet, and/or desktop personal computer, or other computing devices. The computing device further includes an I/O interface connected to devices, such as digital displays, user input devices, etc.

[0092] The functions of the engines may be implemented across distributed computing devices, etc. connected to one another through a communication link. In other examples, functionality of the system may be distributed across any number of devices, including the portable personal computer, smart phone, electronic document, tablet, and desktop personal computer devices shown. The computing device may be communicatively coupled to the network and another network. The networks may be public networks such as the Internet, a private network such as that of a research institution or a corporation, or any combination thereof. Networks can include, local area network (LAN), wide area network (WAN), cellular, satellite, or other network infrastructure, whether wireless or wired. The networks can utilize communications protocols, including packet-based and/or datagram-based protocols such as Internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), or other types of protocols. Moreover, the networks can include a number of devices that facilitate network communications and/or form a hardware basis for the networks, such as switches, routers, gateways, access points (such as a wireless access point as shown), firewalls, base stations, repeaters, backbone devices, etc.

[0093] The computer-readable media may include executable computer-readable code stored thereon for programming a computer (for example, comprising a processor(s) and GPU(s)) to the techniques herein. Examples of such computer-readable storage media include a hard disk, a CD-ROM, digital versatile disks (DVDs), an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. More generally, the processing units of the computing device may represent a CPU-type processing unit, a GPU-type processing unit, a field-programmable gate array (FPGA), another class of digital signal processor (DSP), or other hardware logic components that can be driven by a CPU.

[0094] Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example

configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components or multiple components.

[0095] Additionally, certain aspects are described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In example aspects, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

[0096] In various aspects, a hardware module may be implemented mechanically or electronically. For example, a hardware module may comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a microcontroller, field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC)) to perform certain operations. A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

[0097] Accordingly, the term “hardware module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering aspects in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where the hardware modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different hardware modules at different times. Software may accordingly configure a processor, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

[0098] Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple of such hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connects the hardware modules. In aspects in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware

module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

[0099] The various operations of the example methods described herein can be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example aspects, comprise processor-implemented modules.

[0100] Similarly, the methods or routines described herein may be at least partially processor-implemented. For example, at least some of the operations of a method can be performed by one or more processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but also deployed across a number of machines. In some example aspects, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other aspects the processors may be distributed across a number of locations.

[0101] The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but also deployed across a number of machines. In some example aspects, the one or more processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example aspects, the one or more processors or processor-implemented modules may be distributed across a number of geographic locations.

[0102] Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

[0103] As used herein any reference to “one aspect” or “an aspect” means that a particular element, feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. The appearances of the phrase “in one aspect” in various places in the specification are not necessarily all referring to the same aspect.

[0104] Some aspects may be described using the expression “coupled” and “connected” along with their derivatives. For example, some aspects may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The aspects are not limited in this context.

[0105] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0106] In addition, use of the “a” or “an” are employed to describe elements and components of the aspects herein. This is done merely for convenience and to give a general sense of the description. This description, and the claims that follow, should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0107] This detailed description is to be construed as an example only and does not describe every possible aspect, as describing every possible aspect would be impractical, if not impossible. One could implement numerous alternate aspects, using either current technology or technology developed after the filing date of this application.

What is claimed:

1. A computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, comprising:

receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times;

generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values;

for each sample in the plurality of samples:

computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene;

classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications;

computing, via one or more processors, a confidence score; and

when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample; and

storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

2. The computer-implemented method of claim 1, further comprising:

generating, via one or more processors, a static digital report including at least one classification result.

3. The computer-implemented method of claim 2, wherein generating the static digital report includes censoring a no-call classification result.

4. The computer-implemented method of claim 1, further comprising:

- receiving, via an application programming interface, a request for RNA sequence data;
- processing, via one or more processors, the request using a rules engine to determine an identity of a sender of the request;
- determining, via one or more processors, a role of the identity of the sender of the request;
- configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request;
- generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and
- transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request.

5. The computer-implemented method of claim 4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating censored results when the role of the identity of the sender of the request is a patient.

6. The computer-implemented method of claim 4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating annotated results when the role of the identity of the sender of the request is a clinician.

7. The computer-implemented method of claim 4, wherein the method is performed automatically as part of a bioinformatics pipeline;
- wherein the role of the sender of the request is a programmatic script; and
- wherein generating the modified sequence data by processing the sequence data includes programmatically generating a structured output corresponding to the modified sequence data,
- wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data.

8. The computer-implemented method of claim 1, wherein the number of classifications is two or more.

9. The computer-implemented method of claim 1, further comprising:

- determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

10. The computer-implemented method of claim 9, wherein the alternative therapy includes at least one of: (i) a first-line therapy or (ii) a second-line therapy.

11. The computer-implemented method of claim 9, wherein the alternative therapy includes at least one of:

- (i) a chemotherapeutic drug therapy,
- (ii) a radiation therapy,
- (iii) a surgery therapy,
- (iv) a therapy in a different class,
- (v) a molecular inhibitor therapy,
- (vi) an antibody therapy,
- (vii) a recombinant nucleic acid therapy,
- (viii) an engineered immune cell therapy,
- (ix) a checkpoint inhibitor therapy,
- (x) a cytokine therapy,
- (xi) a cancer treatment vaccine therapy,

(xii) a CAR-cell therapy; or

(xiii) an oncolytic virus therapy.

12. The computer-implemented method of claim 1, wherein the machine learning model is at least one of a Bayesian model, a supervised machine learning model or an unsupervised machine learning model.

13. A computing system, comprising:

- one or more processors; and
 - one or more memories, having stored thereon instructions that when executed, cause the computing system to:
 - receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times;
 - generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values;
- for each sample in the plurality samples:

- compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene;

classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications;

compute, via the one or more processors, a confidence score; and

when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and

store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

14. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to:

- generate, via one or more processors, a static digital report including at least one classification result.

15. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to:

- receive, via an application programming interface, a request for RNA sequence data;
- process, via one or more processors, the request using a rules engine to determine an identity of a sender of the request;
- determine, via one or more processors, a role of the identity of the sender of the request;
- configure, via one or more processors, a rules engine based on the role of the identity of the sender of the request;
- generate, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and
- transmit, via electronic network, the modified sequence data to a computing device of the sender of the request.

16. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to:

- generate censored results when a role of an identity of a sender of a request is a patient.

17. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to:

generate annotated results when a role of an identity of a sender of a request is a clinician.

18. The computing system of claim 13, wherein the number of classifications is two or more.

19. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to:

determine, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

20. A computer-readable media having stored thereon a set of non-transitory computer-readable instructions that, when executed, cause a computer to:

receive, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times;

generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values;

for each sample in the plurality samples:

compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene;

classify, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications;

compute, via one or more processors, a confidence score; and

when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and

store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

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